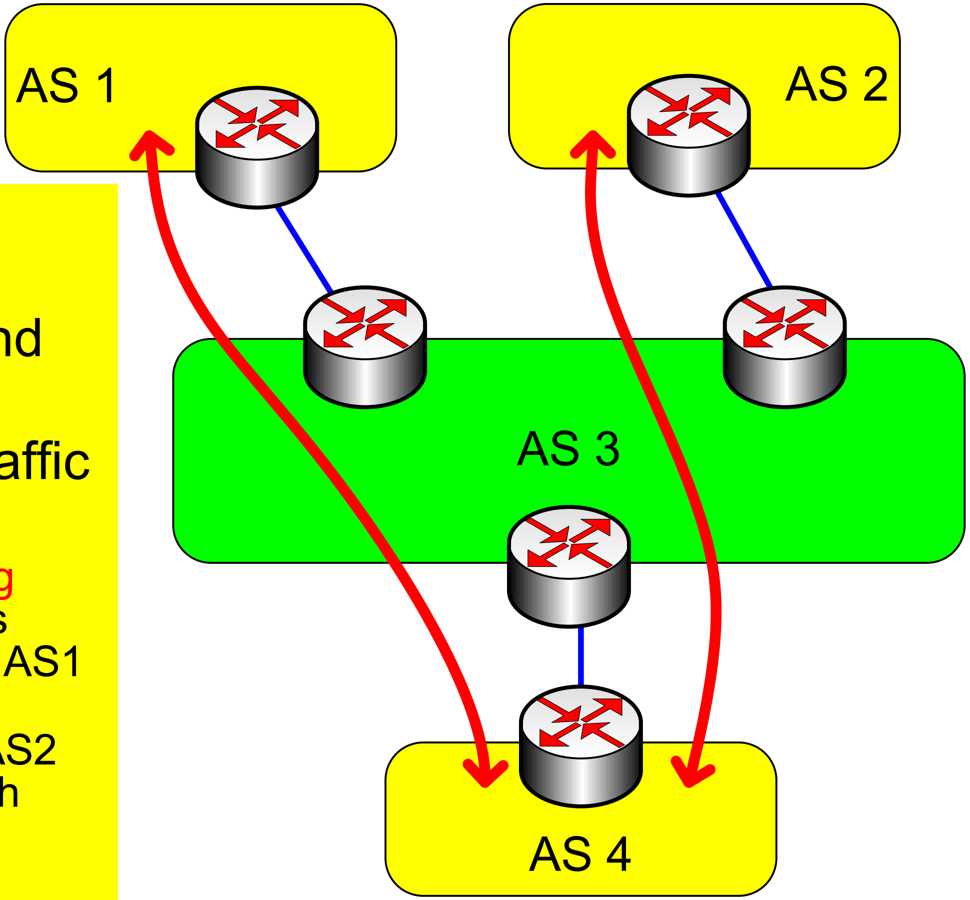
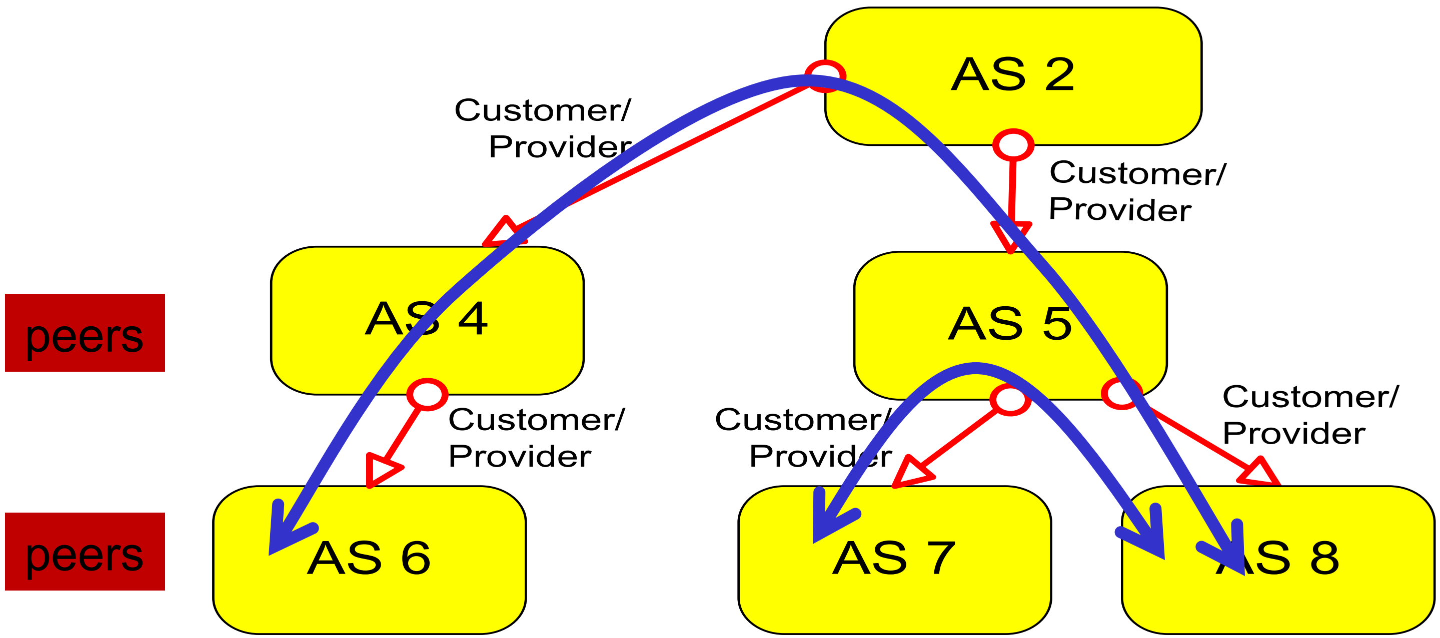
Selective transit

Example:

- AS 3 carries traffic between AS 1 and AS 4 and between AS 2 and AS 4
- But AS 3 does **not** carry traffic between AS 1 and AS 2
  - The example shows a routing policy. In other words, AS3 is perfectly capable of carrying AS1 -> AS2 traffic, but a policy decision prevents AS1 and AS2 from using AS3 to reach each other.

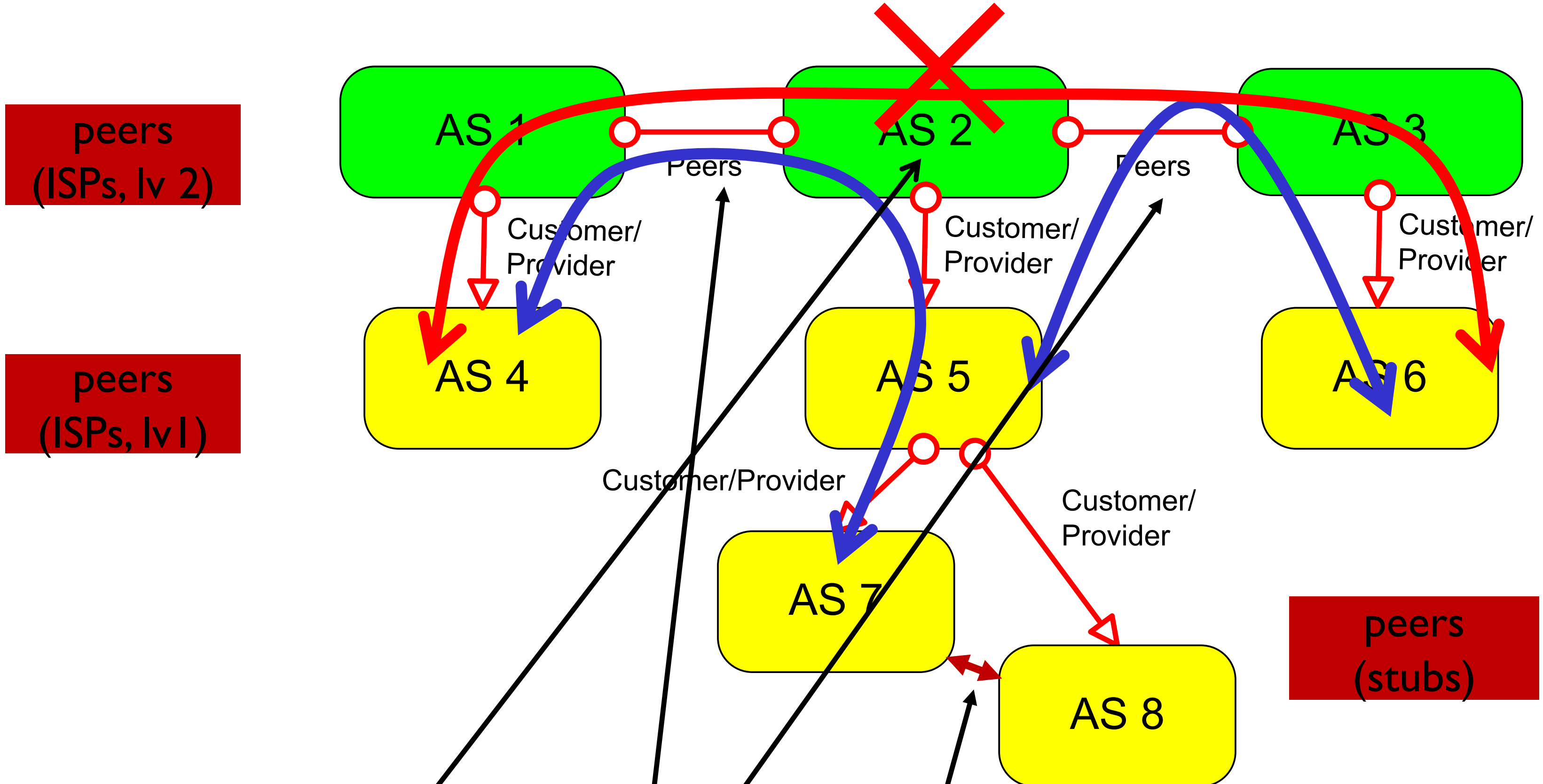


Customer/Provider and Peers



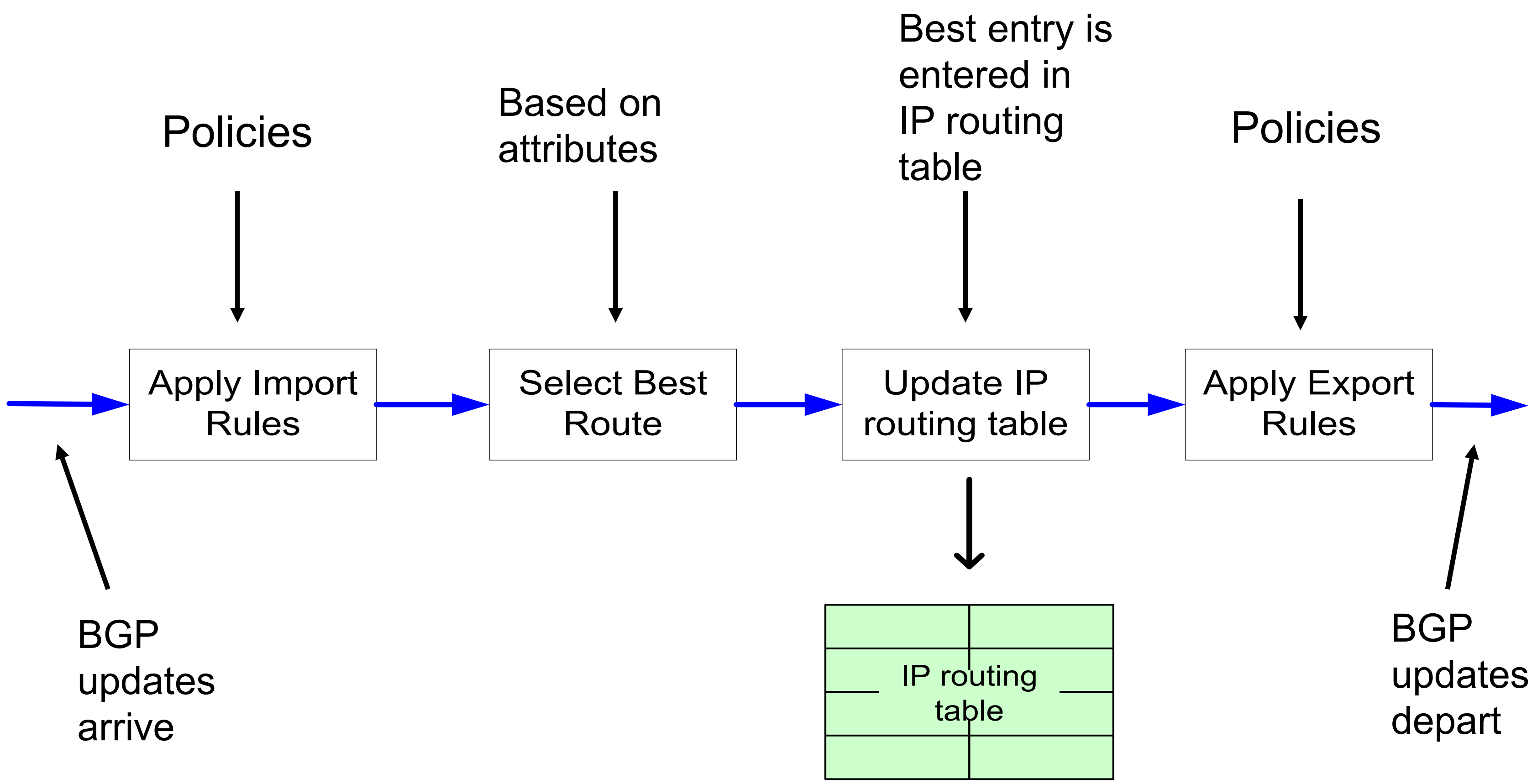
- a stub network typically obtains access to the Internet through a transit network. E.g., AS7 –> AS5 –> AS 8
- a transit network that is a provider may be a customer of another network (provider) – AS4 is a customer of AS2 as is AS5.
- customer pays provider for service

Customer/Provider and Peers



- stubs can have peer relationships – direct link, carries no transit
- transit networks can have a peer relationship
- peers provide transit between their respective customers
- peers **<u>do not</u>** provide **<u>transit</u>** between peers, i.e., traffic from AS1 to AS3 cannot go through AS2.
- peers have to go up one layer to reach another peer if not directly connected
- peers normally do not pay each other for service

Import and Export Policies



Why different Intra-, Inter-AS routing ?

policy:

- inter-AS: manager of an AS wants control over how its traffic is routed externally, and who routes through its net (not applicable for STUB networks).
- intra-AS: single admin, so no policy decisions needed

scale:

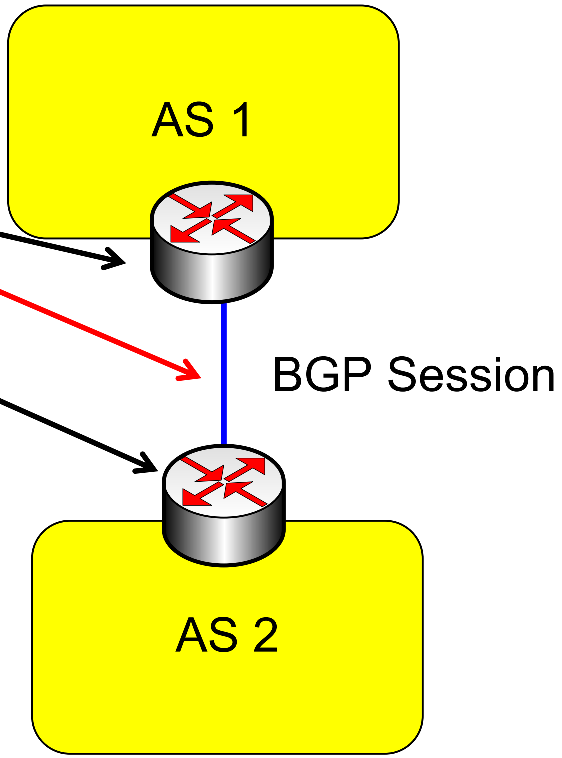
- hierarchical routing saves table size, reduced update traffic

performance:

- intra-AS: can focus on performance (e.g., cost)
- inter-AS: policy may dominate over performance

BGP interactions

- BGP is executed between two routers
  - BGP session
  - BGP peers
- procedure:
  1. *establishes TCP connection (port 175) to BGP peer*
  2. *exchange all BGP routes*
  3. *as long as connection is alive: Periodically send incremental updates*
- **Note:** Not all autonomous systems need to run BGP. On many stub networks, the route to the provider can be statically configured



BGP messages

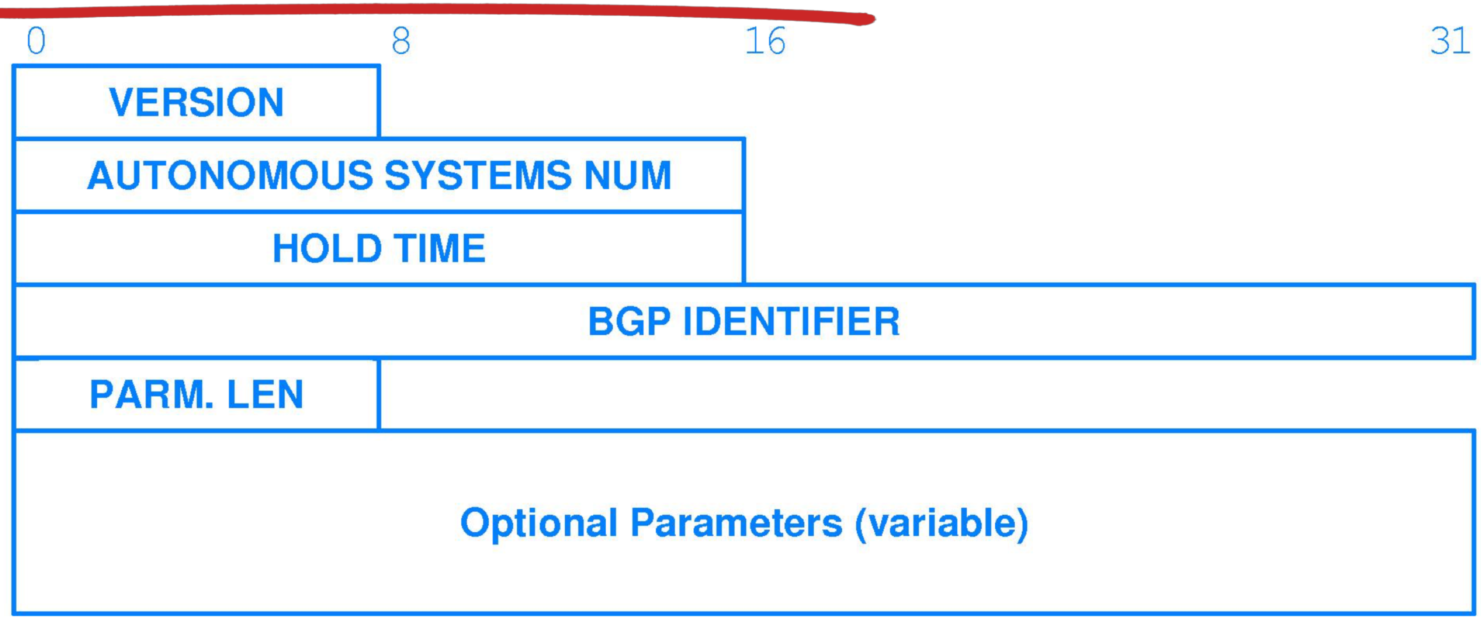
- BGP messages exchanged between peers over TCP session
- BGP messages:
  - OPEN: opens TCP connection to remote BGP peer (port 179) and authenticates sending BGP peer
  - UPDATE: advertises new path (or withdraws old)
  - KEEPALIVE: keeps connection alive in absence of UPDATES; also ACKs OPEN request
  - NOTIFICATION: reports errors in previous msg; also used to close connection

BGP message header

The format of the header that precedes every BGP message.

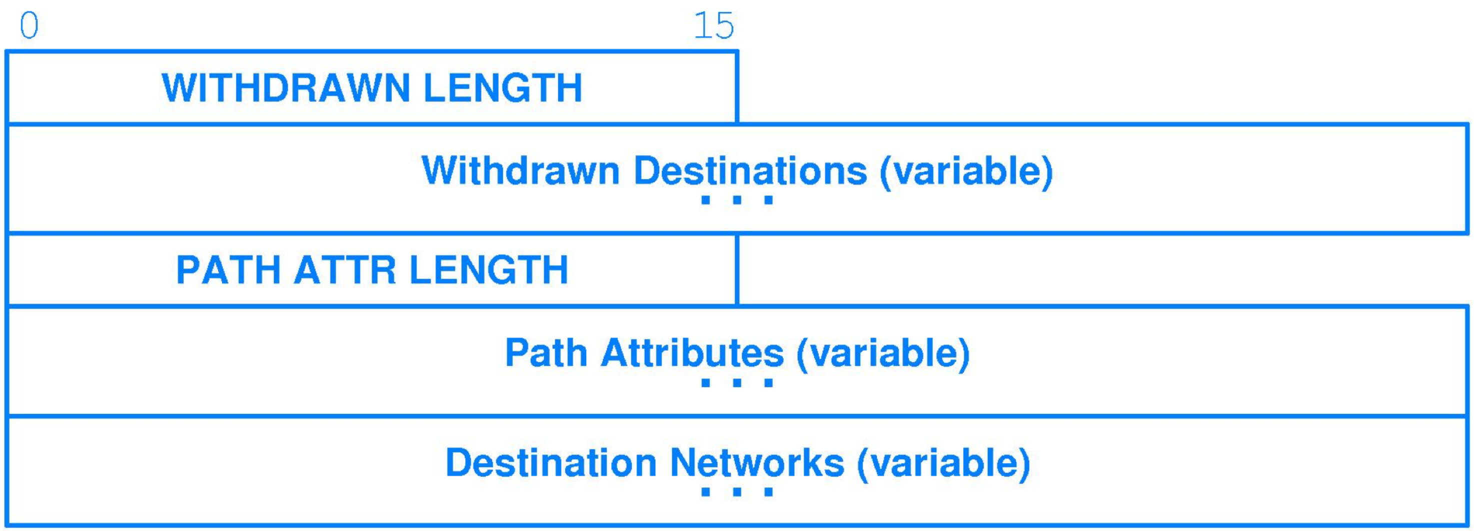
Marker is an agreed upon value (synchronization pattern) between two peers. Usually all one’s, but can be used for authentication. Used to synchronize the two ends.
Length gives total message length in octets
Type contains one of the message types shown in previous slide

BGP Open message



The format of the IPv4 BGP OPEN message that is sent at start-up. Octets shown in the figure follow the standard message header.

BGP Update message



BGP UPDATE message format in which variable size areas of the message may be omitted. These octets follow the standard message header.

Note that any field labeled "variable", can be omitted if there is no information for a parameter

BGP route updates

- BGP route advertisement is sent in a BGP UPDATE message

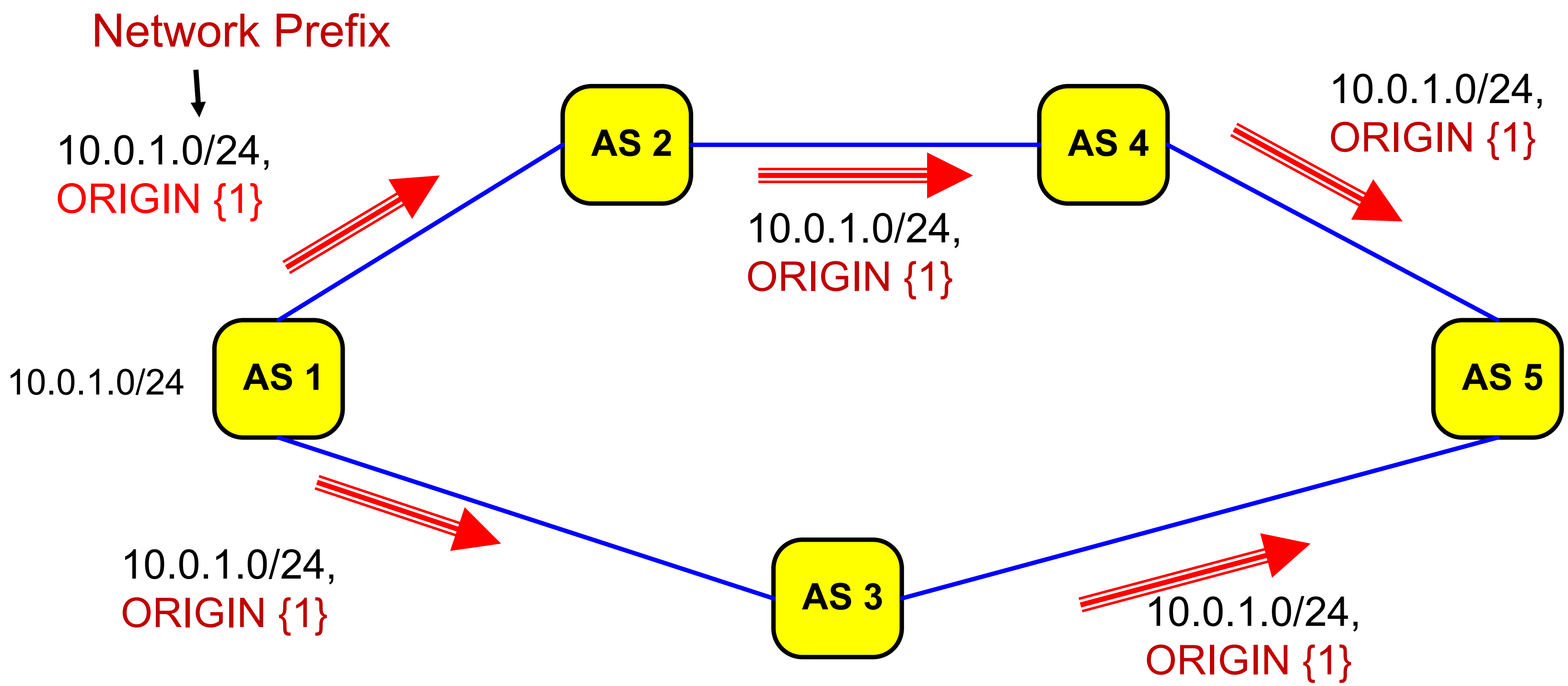
- a route is announced as a **Network Prefix,** e.g., 10.0.1.0/24, and **Attributes**

- **Attributes** specify details about a route:
  - Mandatory attributes:

    ORIGIN
    AS_PATH
    NEXT_HOP
  - many other attributes

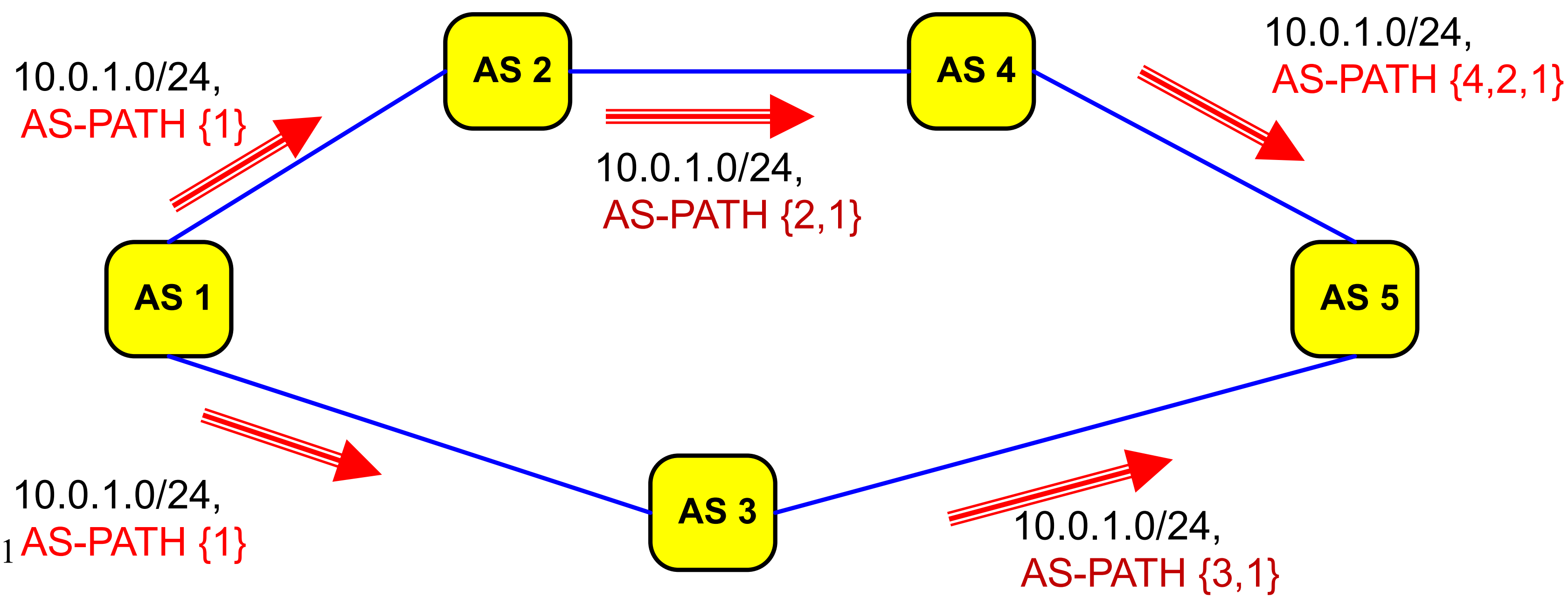
ORIGIN attribute

- originating domain sends a route to a network (here 10.0.1.0/24) with ORIGIN attribute (AS number)



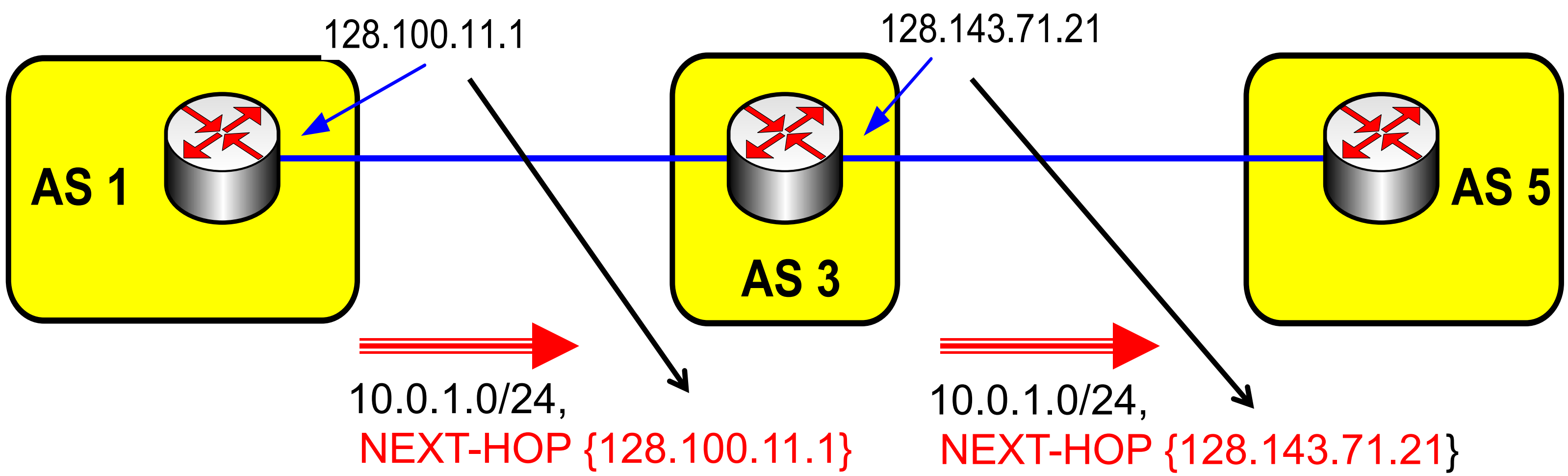
AS-PATH attributes

- each AS that propagates a route **prepends** its own AS number
  - AS-PATH creates a full path to reach the network prefix 10.0.1.0/24
- path information prevents routing loops from occurring
- path information also provides information on the length of a path (no. of ASes enroute, by default, a shorter route is preferred)
- **Note**: BGP aggregates routes according to CIDR rules

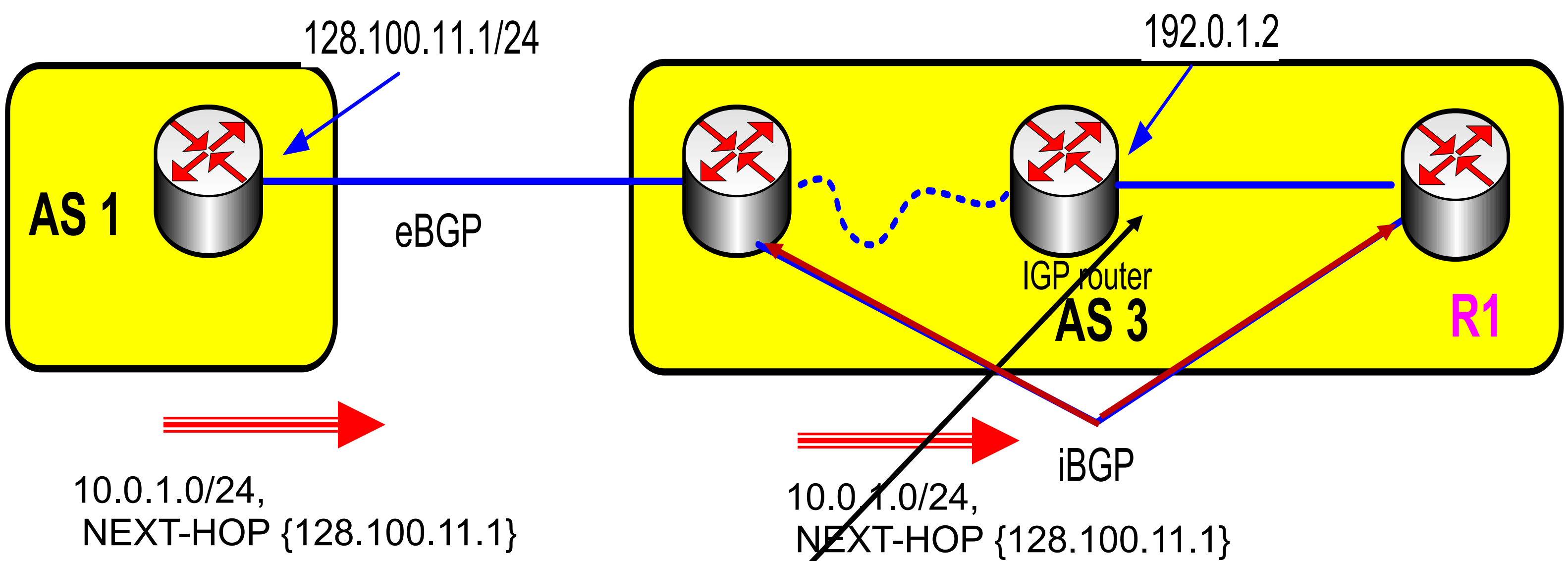


NEXT-HOP attributes

- each router that sends a route advertisement, includes its own IP address of the forwarding port in a NEXT-HOP attribute
- the attribute provides information for the routing table of the receiving router in the next AS on the path



BGP NEXT-HOP -> IGP information



At R1:

IGP Routing table

Dest.	Next hop
128.100.11.0/24	192.0.1.2

BGP info

Dest.	Next hop
10.0.1.0/24	128.100.11.1

Combined Routing table At R1

Dest.	Next hop
128.100.11.0/24	192.0.1.2
10.0.1.0/24	192.0.1.2